

Early Prediction of Diabetes Risk Using Explainable Artificial Intelligence (XAI)

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*A project report submitted in partial fulfillment of the requirements
for B.Sc. degree in Information Technology. Qassim-Saudi Arabia*

1447/1448 (2025/2026)

Certificate

This is to certify that the enclosed project report has been systematically prepared and documented under my direct supervision. The report is hereby approved for submission and academic evaluation.

Dr. Manal Alghaith

Dedication

This project is wholeheartedly dedicated to our families, especially our parents, whose continuous support and encouragement inspired us during difficult times. We remain profoundly grateful for their unwavering moral and emotional care. We also dedicate this endeavor to our esteemed supervisor, whose invaluable guidance illuminated our path toward achieving these academic objectives.

Abdulmajeed, Fahad, Abdullah and Abdulmohsen

Acknowledgments

First and foremost, we express our deepest gratitude to Allah for granting us the strength, knowledge, and perseverance required to reach this milestone. We extend our sincere appreciation to our supervisor, Dr. Manal Alghaith, for her exceptional mentorship, constructive feedback, and continuous support. We also acknowledge the esteemed faculty members of the Information Technology Department for providing valuable insights that significantly enriched our research. Finally, we convey our heartfelt thanks to our parents for their unwavering patience throughout this endeavor, and we highly value the cooperation, commitment, and dedication demonstrated by our team members during all phases of the project.

Abdulmajeed, Fahad, Abdullah and Abdulmohsen

Abstract

This project addresses the challenge of late-detected diabetes by proposing an early prediction system based on Explainable Artificial Intelligence (XAI). Traditional diagnostic approaches are often costly, slow, and inaccessible, while many AI-based solutions lack transparency. To overcome these limitations, the system utilizes the XGBoost algorithm for predictive modeling and integrates SHAP (SHapley Additive exPlanations) to provide transparent predictions using a validated Saudi cross-sectional dataset. The study delivers an explainable decision-support prototype that predicts diabetes risk while highlighting influential demographic, clinical, and lifestyle risk factors. This approach enhances trust and supports preventive healthcare strategies without replacing professional clinical diagnosis. The proposed framework achieved a Recall (Sensitivity) of 96.41%, demonstrating its effectiveness in identifying high-risk diabetes cases while maintaining model interpretability through explainable AI techniques.

Keywords: Explainable AI (XAI), Diabetes Prediction, XGBoost, SHAP, Preventive Healthcare, Saudi Arabia.

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Table 1: List of Abbreviations

Abbreviation	Full Name
ADASYN	Adaptive Synthetic Sampling Approach
AI	Artificial Intelligence
ANOVA	Analysis of Variance
AUC	Area Under the Curve
AutoML	Automated Machine Learning
AUROC	Area Under the Receiver Operating Characteristic curve
BMI	Body Mass Index
CGM	Continuous Glucose Monitoring
CNN	Convolutional Neural Network
CRF	Clinical Risk Factor
DKA	Diabetic Ketoacidosis
DL	Deep Learning
DR	Diabetic Retinopathy
DT	Decision Tree
ECG	Electrocardiogram
EHR	Electronic Health Records
F1-Score	F-Measure (Harmonic mean of precision and recall)
FN	False Negative
FP	False Positive
FPG	Fasting Plasma Glucose
GB	Gradient Boosting
GDM	Gestational Diabetes Mellitus
HbA1c	Glycated Hemoglobin
HIS	Hospital Information System
IEEE	Institute of Electrical and Electronics Engineers
IRB	Institutional Review Board
KAU	King Abdulaziz University
KSA	Kingdom of Saudi Arabia
LIME	Local Interpretable Model-agnostic Explanations
LR	Logistic Regression
LSTM	Long Short-Term Memory
ML	Machine Learning
NRI	Normalized Relative Influence
OGTT	Oral Glucose Tolerance Test
PCA	Principal Component Analysis
PIDD	Pima Indians Diabetes Dataset
RF	Random Forest
RFE	Recursive Feature Elimination
SGB	Stochastic Gradient Boosting
SH	Severe Hypoglycemia
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machines
T2DM	Type 2 Diabetes Mellitus

Table 1: List of Abbreviations (Continued)

Abbreviation	Full Name
TN	True Negative
TP	True Positive
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Chapter 1

INTRODUCTION

INTRODUCTION

1.1 Overview

Artificial Intelligence (AI) and Machine Learning (ML) are fundamental to modern computational systems. These technologies enable automated data processing and pattern recognition. They also power advanced predictive analytics across various sectors. However, high-performing AI models often operate as “black boxes”. In these models, the internal reasoning behind outputs remains non-transparent. This lack of interpretability can limit trust and accountability. It also hinders adoption in sensitive application domains.

To address this limitation, Explainable Artificial Intelligence (XAI) has emerged as a specialized research field. XAI develops techniques that provide high predictive performance alongside clear explanations. These methods identify how input features contribute to specific predictions. Interpretability techniques are generally categorized into intrinsic and post-hoc explainability. Intrinsic models, such as decision trees, are interpretable by design. However, they may underperform with complex datasets. Post-hoc approaches enable the interpretation of complex models after training. Examples include Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). Such transparency is critical for fostering trust and supporting model validation. It also ensures alignment with ethical and regulatory requirements.

In the medical domain, interpretability is particularly essential because decisions directly affect patient health outcomes. Diabetes, a chronic metabolic disorder with high prevalence in Saudi Arabia, reveals an alarming and accelerating trend in the burden of Type 2 Diabetes Mellitus (T2DM). Epidemiological modeling suggests that the prevalence rate of T2DM among the Saudi adult population (aged ≥ 25 years) increased significantly, rising from 8.5% in 1992 to approximately 17.7% around 2000, and further to an estimated 39.5% by 2022 [1].

The age-standardized prevalence among adults (20–79 years) was 19.6% in 2011, rose to 23.1% in 2024, and is projected to reach 25.4% by 2050 [2]. This substantial and sustained increase in prevalence underscores that diabetes is a major, escalating health problem in KSA, requiring early detection and preventive intervention to reduce long-term complications. Therefore, predictive models used in this context must not only be accurate but also explainable to healthcare professionals, patients, and policymakers.

In this project, Explainable Artificial Intelligence is employed to develop an early prediction model for diabetes risk tailored to the Saudi Arabian population. The objective is to combine predictive performance with interpretability, ensuring that the reasoning behind predictions is clear, clinically meaningful, and supportive of informed healthcare decision-making.

1.2 Project Scope

The scope of this project primarily focuses on the medical and healthcare domains. Early detection of diabetes plays a vital role in reducing morbidity. It also improves overall patient outcomes. The proposed system aims to support healthcare professionals, such as physicians and nurses. It provides a reliable AI model to identify high-risk individuals at an early stage. This predictive capability assists in clinical decision-making and screening processes. It also facilitates preventive healthcare strategies to reduce medical and economic burdens.

The project also extends its scope to include the general public. This includes individuals concerned about their susceptibility to diabetes. The system offers a simple and accessible prediction tool for self-assessment. Users can gain insights into the key factors influencing their potential risk. This is especially beneficial for individuals with a family history of diabetes. It also serves those with predisposing conditions such as obesity or hypertension.

The integration of XAI ensures the system does not function as a “black box”. It provides transparent and interpretable reasoning behind each prediction. SHAP was selected as the primary XAI framework, while LIME remains a candidate for future comparative analysis. Practitioners can understand which features influence the AI model’s decision. This interpretability enhances trust and accountability in healthcare applications.

In summary, the scope covers both clinical and non-clinical environments. The system aims to bridge the gap between medical technology and public accessibility. It contributes to early awareness and encourages proactive health management. The proposed system is intended solely as a decision-support and risk assessment tool. It is not designed to replace professional clinical diagnosis or medical judgment.

1.3 Problem Statement

Diabetes mellitus is one of the most prevalent chronic diseases worldwide. Millions of individuals remain undiagnosed due to a lack of early symptoms. Limited access to regular medical screening

further contributes to this issue. Unawareness often results in delayed diagnosis. This delay leads to severe complications such as cardiovascular disease and kidney failure.

Early identification of high-risk individuals is therefore a critical public health necessity. Traditional diagnostic methods rely heavily on laboratory tests and clinical expertise. These processes can be time-consuming, costly, and inaccessible to some populations. AI models have demonstrated high potential in predicting diabetes risk. However, their “black-box” nature limits clinical trust and adoption. Healthcare professionals require transparent and interpretable decision-making processes.

To address this issue, our project employs XAI for early diabetes risk prediction. XAI enhances transparency by providing human-understandable explanations. It shows how features like BMI and glucose levels influence predictions. Unlike conventional AI, XAI combines accuracy with accountability. It allows users to understand why and how a particular prediction is made. This is achieved through tools such as SHAP. This framework provides both global and local explanations of model behavior.

By integrating XAI, our project aims to bridge the gap between ML performance and clinical usability. This approach enhances diagnostic transparency. It also supports informed medical decision-making. Ultimately, it contributes to early detection and preventive healthcare strategies.

1.4 Primary Goal

The primary goal is to establish a robust AI-driven framework for early diabetes identification. This framework is specifically tailored to the population in KSA. The system identifies individuals at high risk while providing transparent reasoning. This approach supports informed clinical decision-making. It also enhances user trust in automated health assessment tools. The project aligns with national healthcare priorities under Saudi Vision 2030. It seeks to demonstrate how XAI can improve prevention strategies in real-world contexts. Finally, it aims to advance patient-centered care within the Saudi healthcare system.

1.5 Aim and Objectives

The project is structured around technical milestones to translate this vision into a prototype.

These objectives bridge theoretical XAI concepts with practical healthcare applications:

- **Review existing literature** to identify current advancements and research gaps in explainable diabetes models.
- **Obtain accurate and trustworthy data** from reliable sources to ensure a robust foundation for development.
- **Develop a predictive model** based on established ML techniques to ensure high diagnostic performance.
- **Implement interpretability layers** using SHAP to make the decision-making process transparent.
- **Evaluate the system's effectiveness** using appropriate metrics to measure reliability and practical utility.

1.6 Study Plan

A comprehensive Study Plan has been developed to ensure successful project completion. The schedule spans from October 2025 to March 2026. The timeline is divided into sequential phases to ensure systematic progress.

The first phase covers the Introduction and Literature Review (October - November). This is followed by Methodology design and Data Collection (November - December). The Implementation and Testing phase takes place from December to February. The project concludes with Results analysis and the Conclusion in February and March. The detailed task distribution is illustrated in the Gantt Chart in Figure 1.1.

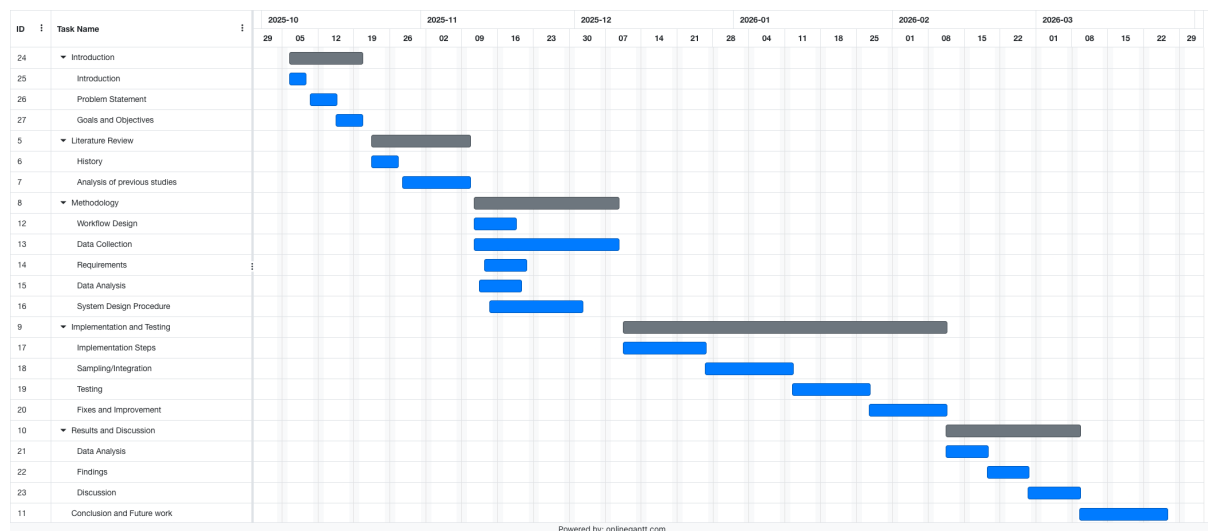


Figure 1.1: Project Study Plan and Gantt Chart.

Chapter 2

LITERATURE REVIEW

LITERATURE REVIEW

2.1 Background

Diabetes mellitus represents a global metabolic crisis characterized by persistent hyperglycemia. This condition stems from impaired insulin secretion or systemic insulin resistance. Its progression is linked to severe complications, including cardiovascular failure and vision-threatening DR. Standard clinical diagnosis relies on FPG, OGTT, and HbA1c measurements. However, these tools often identify the disease only after significant metabolic dysregulation occurs. This diagnostic delay restricts the window for effective preventive intervention.

The digitalization of healthcare has led to an explosion of structured medical data. Massive datasets are now available through EHR, CGM, and various biomarkers. This data wealth enabled the integration of AI and ML in predictive medicine. Advanced algorithms can now identify high-risk groups long before clinical manifestation. Despite their accuracy, many high-performing models function as opaque “black boxes.” The lack of transparent reasoning hinders clinician trust and limits real-world deployment.

To bridge this gap, XAI has emerged as a transformative paradigm in medical informatics. XAI aims to provide high predictive performance alongside human-understandable explanations. Frameworks such as SHAP and LIME enable both local and global interpretability. These tools highlight critical features like BMI, age, and glucose levels in each prediction. Such transparency supports evidence-based decision-making and ensures clinical accountability. Modern research now favors hybrid pipelines that combine advanced learners with XAI mechanisms. This evolution is essential for building ethically sound and patient-centered diabetes care systems.

2.2 Review of Existing Studies

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have significantly transformed diabetes prediction and management, with a growing emphasis on Explainable Artificial Intelligence (XAI) to enhance transparency, interpretability, and clinical trust. The integration of XAI tools—such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations)—has become increasingly essential for translating predictive insights into clinically meaningful interpretations. This literature review synthesizes twenty key studies that explore the intersection of high-performance AI models and interpretability frameworks, underscoring their combined potential to improve diagnostic accuracy, transparency, and healthcare decision-making.

Initial research efforts have focused on optimizing classical ML algorithms for diabetes prediction while addressing common data-related challenges. Isfazzaman Tasin et al. [3] developed a diabetes prediction framework that integrated public and private datasets while resolving class imbalance using SMOTE and ADASYN techniques. Among various algorithms, XGBoost combined with ADASYN achieved the highest accuracy (81%), and SHAP/LIME analyses identified glucose, BMI, age, and insulin as primary predictors, illustrating the potential of interpretable classical ML when coupled with balanced data preprocessing.

Similarly, Raza Hasan et al. [4] applied an Automated Machine Learning (AutoML) approach (AutoGluon) integrated with SHAP, LIME, Integrated Gradients, Attention, and Counterfactual Analysis to mitigate the “black box” challenge. Their model achieved 76.6% test accuracy and generalized well across datasets (average accuracy $\approx 85\%$), with glucose and BMI consistently emerging as critical variables.

Mustafa Kutlu et al. [5] extended this exploration using SHAP analysis with an XGBoost model trained on a large-scale health dataset of over 250,000 individuals. Although their model reached high accuracy (86.6%), its low recall ($\approx 17\%$) indicated persistent class imbalance. SHAP identified General Health Status, High Blood Pressure, Age, and BMI as leading predictors, highlighting interpretability’s value in diagnosing model weaknesses. In a longitudinal context, Lara Lama et al. [6] employed interpretable ML models with SHAP to predict diabetes risk over 10–20 years in a Swedish cohort. Their findings revealed BMI, waist-hip ratio, blood pressure, age, and family history as key long-term predictors, demonstrating XAI’s utility for preventive healthcare strategies. Complementarily, Hung Viet

Nguyen et al. [7] proposed a deep learning-based hybrid model (HyperTab-LIME) combining hypernetwork architectures with LIME for prediabetes prediction in men aged 30 and above. The model achieved a strong AUC of 0.8429, confirming that deep learning can be both accurate and explainable when supported by XAI frameworks.

Beyond predictive accuracy, interpretability was further enhanced through hybrid and ensemble approaches. Srinivasu et al. [8] introduced an explainable AI system utilizing wearable sensor data (spectrogram images) for early diabetes detection. Their CNN-Bi-LSTM model achieved 96.4% accuracy and leveraged SHAP to visualize influential features. However, the study's limitations included a small dataset (6,735 samples) and a lack of clinical validation. Bahad et al. [9] proposed an explainable ensemble model using AdaBoost, XGBoost, and Random Forest algorithms. With SHAP and LIME providing transparent explanations, the model achieved 85.8% accuracy, 92% sensitivity, and 88% specificity on the Pima Indians Diabetes Dataset (PIDD), confirming glucose, BMI, and insulin as top contributors. Similarly, Uysal [10] compared multiple ML algorithms, identifying SVM (82% accuracy) and Random Forest (81% accuracy) as the most successful models. The study then used SHAP extensively to provide a detailed interpretability analysis, identifying Glucose, Age, and BMI as the most significant predictors. Vinh and Byeon [11], in a literature review, synthesized research on transparent diabetes prediction, highlighting studies that integrated models like Gradient Boosting with XAI tools such as SHAP and LIME to improve transparency. The paper emphasized that XAI successfully identified key factors like glucose and BMI but also noted that reliance on single datasets (like PIDD) and a lack of clinical validation constrained the applicability of many existing models. Ganguly and Singh [12] further enhanced interpretability using an ensemble XAI framework combining Random Forest, Decision Tree, and SVM. SHAP and LIME analyses highlighted glucose, BMI, and age as major predictors, yet dataset homogeneity (female-only) restricted the model's generalizability.

In addition to diabetes onset prediction, XAI has been applied to detect and manage complications such as diabetic retinopathy (DR). Rathinam et al. [13] proposed a hybrid I²WD-DMIC model combining an Improved Intelligent Water Drop optimization algorithm with a discriminated multi-instance classification approach, which achieved strong detection performance ($\approx 99.8\%$ accuracy) and computational efficiency. However, limited dataset transparency and overcomplex optimization techniques reduced interpretability. Another study by Sad-Houari et al. [14] developed an automatic diabetes-prediction system using a dataset

of 823 Algerian patients, incorporating ANOVA and Recursive Feature Elimination (RFE) for feature selection. While the study effectively identified risk factors, it achieved high quantitative performance, with several models (SVM, KNN, LR) reaching 97.60% accuracy. Extending these efforts, Mohsen and Shah [15] introduced ECG-DiaNet, a deep learning model integrating ECG signals with clinical risk factors (CRFs) using Qatar Biobank data. The model achieved an AUROC of 0.845, outperforming CRF-only models (AUROC = 0.8217), although limited external validation hindered broader application. Similarly, Wändell et al. [16] developed a Stochastic Gradient Boosting (SGB) model within a Primary Health Care framework that identified hypertension and obesity as primary predictors of diabetes onset. However, reliance on administrative data and missing lifestyle variables weakened predictive robustness. Khanna et al. [17] focused on Gestational Diabetes Mellitus (GDM) and developed an ensemble stack model with SMOTE-ENN data balancing, achieving a high accuracy of 96%. This study stood out for its comprehensive use of XAI, employing SHAP, LIME, ELI5, and Anchor to validate the model's predictions.

To address class imbalance and interpretability challenges, Shaheen et al. [18] developed two ensemble deep-learning models—Hi-Le (Highway + LeNet) and HiTCLe (Highway + TCN + LeNet)—on an imbalanced dataset corrected via ProWSyn. The Hi-Le model achieved 94% accuracy with an F1 score of 96%, and SHAP revealed glucose and HbA1c as dominant predictors. Despite high accuracy, heavy computational requirements and risk of overfitting due to oversampling limited real-world deployment. Ahmed et al. [19] prioritized interpretability by training a Logistic Regression (LR) model on CDC BRFSS 2015 data, explained through SHAP and LIME. Achieving 86% accuracy but low recall (0.15) for diabetic cases, the study highlighted persistent class imbalance issues while offering valuable local/global explanations emphasizing high blood pressure, general health, and BMI. Khokhar et al. [20] conducted a systematic review outlining typical accuracy ranges (80–90%) for classical ML methods, identifying recurring issues such as imbalance, “black-box” models, and insufficient validation. The review underscored the importance of XAI integration and standardized evaluation metrics. Parab et al. [21] extended this perspective by reviewing AI in clinical diabetes care, stressing that models must be explainable, externally validated, and seamlessly integrated into healthcare workflows to gain clinical acceptance. Finally, Ihalapathirana et al. [22] developed an explainable boosting model to predict severe hypoglycemia (SH) and diabetic ketoacidosis (DKA) in Type 1 Diabetes. Using XGBoost, LightGBM, and CatBoost with SHAP, they achieved F1 scores of 84–85% for

SH and 79% for DKA, offering one of the most balanced trade-offs between accuracy, fairness, and interpretability, albeit without external validation.

2.3 Comparative Analysis

Table 2.1 provides a comparative analysis of twenty academic studies focused on AI for diabetes prediction, with a significant emphasis on model interpretability. A review of these studies reveals several key trends, methodological choices, and performance outcomes.

Table 2.1: Accuracy-Based Comparative Analysis of AI Models for Diabetes Prediction

Ref	Technique(s) Used	Accuracy
[13]	• I²WD-DMIC model on retinal images (for DR). • Used I²WD (feature selection).	≈99.8%
[14]	• SVM, KNN, DT, & LR. • Used RFE & ANOVA for feature selection.	97.60%
[8]	• A hybrid deep learning model combining CNN and Bi-LSTM. • Used SHAP for interpretability.	96.4%
[17]	• Ensemble Stack (with SMOTE-ENN data balancing). • Used SHAP, LIME, ELI5, Anchor & other XAI tools.	96%
[18]	• Ensemble DL models (Hi-Le and HiTCLe). • Used SHAP for interpretability.	94%
[11]	• A comparison of ML models, with Gradient Boosting (GB) performing best. • Used LIME and SHAP for interpretability.	90%
[5]	• XGBoost model. • Used SHAP for interpretability.	86.6%
[19]	• LR and RF models. • Used LIME and SHAP for interpretability.	86%
[9]	• Ensemble ML models, with XGBoost performing best. • Used LIME and SHAP for interpretability.	85.8%
[4]	• AutoML (AutoGluon) ensemble model. • Used SHAP, LIME, Integrated Gradients, and others.	85.01%
[22]	• XGBoost, LightGBM, CatBoost, and other models. • Used SHAP for interpretability.	84.95% (F1)
[15]	• ECG-DiaNet (Multimodal Neural Network). • (XAI was recommended for future work).	0.845 (AUC)
[10]	• A comparison of ML models, with Random Forest (RF) performing best. • Used SHAP for interpretability.	(81-82%)
[3]	• XGBoost classifier with ADASYN. • Used LIME and SHAP for interpretability.	81%
[7]	• A hybrid deep learning model (HyperTab). • Used LIME for interpretability.	76.86%
[16]	• Stochastic Gradient Boosting (SGB). • Used NRI for feature importance.	0.736 (AUC)

**Note: Studies [1], [2], [6], [12], [20], and [21] were excluded from the comparison table because they were review papers, longitudinal studies, or did not report standardized evaluation metrics suitable for direct comparison.*

2.4 Summary and Discussion

1. **Wide Performance Variance:** The table shows a clear lack of a single performance benchmark. Reported accuracy varies dramatically across the reviewed studies. Results range from exceptional figures like 99.8% and 97.6% to a more common cluster between 76% and 87%.

2. **Emphasis on Explainability (XAI):** The table confirms that model transparency is now a central goal of modern research. The vast majority of listed studies explicitly integrate XAI tools. SHAP is the most dominant technique used for global and local feature importance. LIME is also frequently used to explain individual patient predictions.

3. **Methodological Diversity:** The table illustrates that there is no single consensus on the best technique. High-performing models include specialized DL hybrids and ensemble methods. Other successful approaches pair traditional ML with robust feature selection.

Collectively, these twenty studies demonstrate a consistent evolution in the field. Research is moving from traditional ML models toward explainable and hybrid architectures. These systems aim to balance high performance with clinical interpretability. SHAP and LIME have emerged as dominant XAI tools across diverse methodologies. These tools enable clinicians to understand feature contributions and model rationale. Despite promising results, several persistent challenges remain. These include class imbalance and limited dataset diversity. Many studies still rely heavily on retrospective or single-institution data. There is also a notable lack of clinical and external validation. Future research should prioritize the use of standardized datasets. Real-time clinical trials are necessary for further development. Integration of multimodal data, such as wearable sensors and imaging, will enhance generalizability. Ultimately, XAI holds substantial potential to bridge the gap between performance and clinical trust. This paves the way for ethically sound, transparent, and patient-centered diabetes care systems.

Chapter 3

METHODOLOGY

METHODOLOGY

3.1 Introduction

This chapter outlines the research design, data collection, and analysis methods used to predict diabetes risk using XAI. It also highlights the validation approach and ethical considerations guiding the study.

3.2 Type of Study

This project employs a hybrid methodological approach that integrates two distinct types of inquiry: theoretical investigation and prototype-based development.

The theoretical component involves an analytical study of XAI principles and techniques. This includes a review and synthesis of existing XAI frameworks, an investigation into the mathematical and logical foundations of interpretability methods (such as SHAP), and the conceptual design of a novel framework suited for medical risk prediction. The goal of this component is to establish a robust theoretical grounding for the system.

The prototype-based component translates this theoretical framework into a practical, applied solution. This follows a constructive research paradigm, focusing on the design, implementation, and empirical validation of a software prototype. This involves data preprocessing, the engineering of ML models for diabetes prediction, and the integration of the selected XAI techniques to generate explanations.

This dual approach is essential for the project's objectives. The theoretical work provides the scientific and conceptual rigor, while the prototype development serves as a tangible proof-of-concept, demonstrating the feasibility and utility of the proposed XAI framework in a real-world context.

3.3 Data Collection

3.3.1 Rationale for Data Source Selection

Accessing primary clinical data directly from healthcare institutions in Saudi Arabia presents significant challenges due to rigorous IRB requirements and strict data privacy regulations governing patient records. To navigate these systemic barriers effectively and ensure the highest

standards of data integrity, this study utilizes a validated dataset derived from a published and peer-reviewed scientific research paper. This strategic approach guarantees the use of reliable, ethically cleared data that accurately represents the demographic characteristics of the Western Region of Saudi Arabia, providing a robust foundation for the proposed XAI framework.

3.3.2 Study Design and Setting

This study conducts a secondary analysis of a previously published cross-sectional dataset. The original study, from which the data was sourced, was conducted at King Abdulaziz University (KAU) [23]. Ethical approval and participant recruitment procedures were performed by the original researchers as described in the source publication.

3.3.3 Data Acquisition Method

According to the original study from which this dataset was obtained, the data was collected using an online fill-in form distributed via e-mail to the target population at KAU. The original researchers employed Internet-based technologies to ensure a cost-effective and rapid collection of the large dataset. The questionnaire consisted of closed-ended research questions designed to identify participants at high risk of diabetes based on widely recognized attributes.

3.3.4 Dataset Description

The collected dataset comprises a total of 4,896 subjects. The distribution of the target class is as follows:

- **Diabetic Cases (High Risk):** 990 participants.
- **Non-Diabetic Cases (Low Risk):** 3,906 participants.

The dataset is publicly available on the IEEE DataPort repository (Cross-sectional Type 2 Diabetes Survey Saudi Arabia Western Province) [23].

3.3.5 Feature Description and Encoding

The dataset includes specific predictors (explanatory variables) covering demographic information, physical measurements, and lifestyle habits. The features were encoded for analysis as follows:

1. Demographic Data

- **Gender:** Encoded as Female = 0, Male = 1.
- **Age:** Categorized into four groups:
 - 0: < 40 Years
 - 1: 40 – 49 Years
 - 2: 50 – 59 Years
 - 3: > 60 Years

2. Physical & Clinical Measurements

- **BMI:** Calculated from height and weight (kg/m^2) and categorized as:
 - 0: < 25 kg/m^2
 - 1: 25 – 30 kg/m^2
 - 2: > 30 kg/m^2
- **Waist Size:** Categorized based on gender-specific thresholds (Male/Female) into three levels (0, 1, 2) indicating normal, elevated, or high risk.

3. Lifestyle & Health History

- **Physical Activity:** Defined as engaging in at least 30 minutes of daily exercise (Yes = 0, No = 1).
- **Diet (Fruits, Veg):** Frequency of consumption (Every day = 0, Not every day = 1).
- **Hypertension Medication:** Taking blood pressure medication (No = 0, Yes = 1).
- **Smoking Habits:** (Non-smoker = 0, Smoker = 1).
- **Family History of Diabetes:**
 - 0: No family history.
 - 1: Yes (Grandparents).
 - 2: Yes (Parents).

4. Target Variable (Response)

- **Diagnosis:** Based on exposure to fasting plasma glucose (≥ 5.6 mmol/L) or history of high blood glucose.
 - **High Risk (Diabetic):** Response “YES”.
 - **Low Risk (Non-Diabetic):** Response “NO”.

3.4 Data Analysis

The data analysis phase is a critical component of this study, as it focuses on preparing and structuring the dataset for effective model development and evaluation. At this stage of the project, the analysis procedures are planned and designed to ensure data quality, reliability, and suitability for ML modeling.

The initial steps include data cleaning, which may involve handling missing values, correcting inconsistencies, and removing duplicate or irrelevant records. If multiple data sources are involved, data integration techniques may be applied to combine relevant demographic, clinical, and lifestyle attributes into a unified dataset.

Feature selection methods are planned to identify the most influential variables contributing to diabetes risk prediction, thereby improving model interpretability and reducing computational complexity. Additionally, data transformation techniques such as normalization, scaling, and categorical encoding will be considered where appropriate.

Additionally, to address the class imbalance observed between diabetic and non-diabetic cases, the native cost-sensitive learning capability of the XGBoost algorithm is utilized. Specifically, the `scale_pos_weight` hyperparameter will be tuned to assign higher importance to the positive class (diabetic cases). Since the dataset had already been pre-balanced using SMOTE by the original data collectors [23], additional synthetic oversampling was intentionally avoided to preserve the clinical integrity of the processed dataset.

To address potential dimensionality issues, dimensionality reduction techniques, such as PCA, may be applied if required, depending on feature correlation and model performance. The decision to apply such techniques will be determined during the implementation phase.

Finally, the dataset will be divided into training, validation, and testing subsets (e.g., 70%, 15%, and 15%) to ensure unbiased model evaluation and to minimize the risk of overfitting.

3.5 System Design Procedure

The system design follows a structured pipeline comprising three main processing phases, initiated by fetching raw data from a validated survey-based dataset.

- 1. Data Processing and Splitting Phase:** The raw data undergoes rigorous preprocessing, including cleaning, integration, feature selection, and encoding. As depicted in the system design, the processed data is split into three distinct subsets to ensure robust model development: 70% for training, 15% for validation, and 15% for testing.
- 2. AI Modeling and Explainability Phase:** This phase utilizes the Training Set to train the XGBoost classifier. Simultaneously, the Validation Set is employed in a feedback loop for hyperparameter tuning to optimize the model's performance. Once trained, the model is integrated with the SHAP explainer. SHAP is selected as the primary interpretation framework due to its ability to provide consistent global and local explanations suitable for tree-based models like XGBoost, thereby mitigating the “black-box” nature of the algorithm.
- 3. Validation and Evaluation Phase:** The unseen Testing Set is fed into the performance evaluation module to calculate metrics such as accuracy, recall, and F1-score. Concurrently, the explanations generated by SHAP undergo clinical validation by medical experts to ensure the diagnosis is medically sound.

The detailed architecture, data flow, and interaction between these layers are illustrated in Figure 3.1.

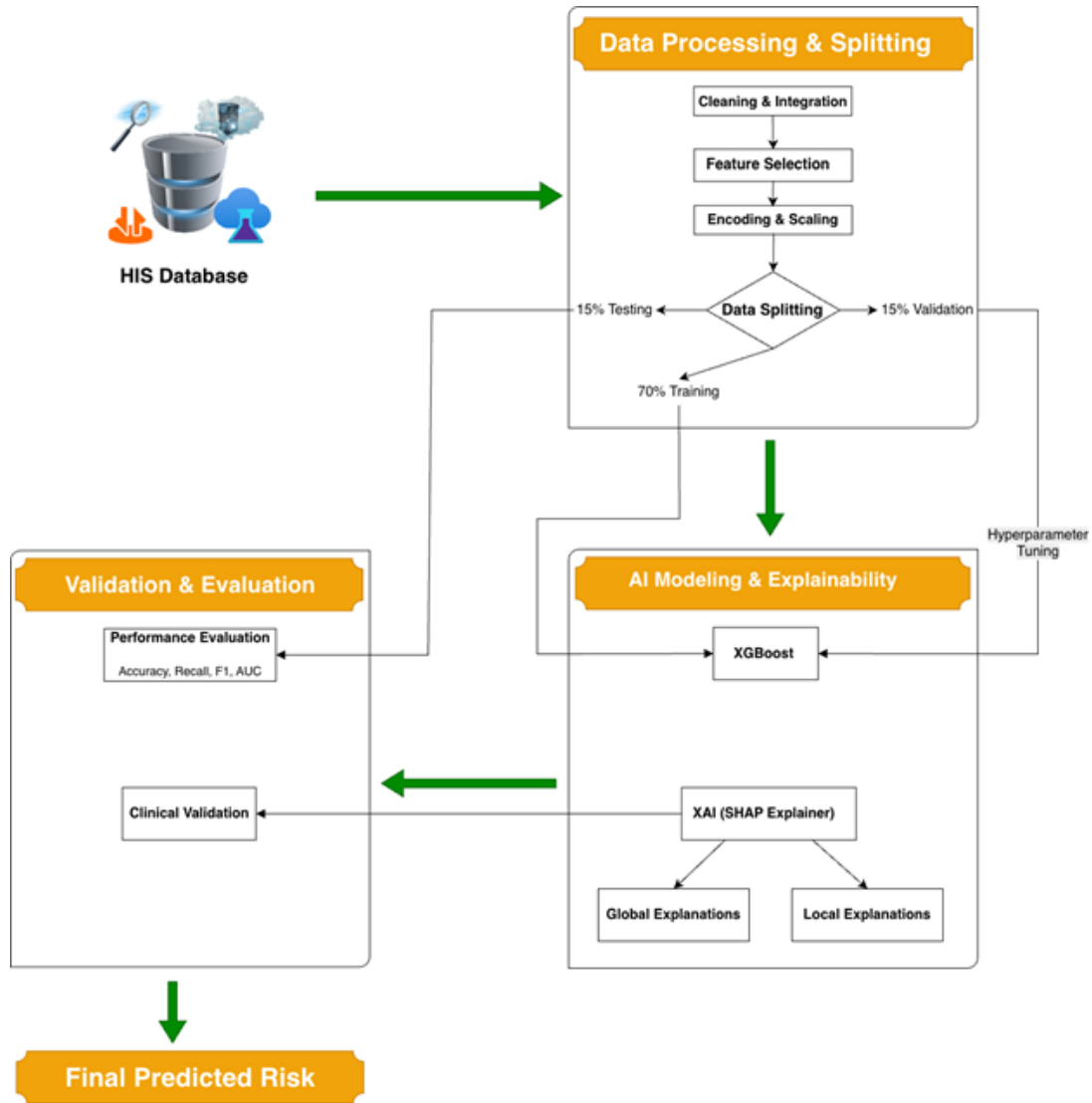


Figure 3.1: Proposed System Architecture for Diabetes Prediction using XAI.

3.6 Summary

In this chapter, the methodological framework for the proposed diabetes prediction system was established. A hybrid approach combining theoretical research and prototype development was adopted to ensure both scientific rigor and practical applicability. The study utilizes a validated cross-sectional dataset from the Western Region of Saudi Arabia, comprising 4,896 subjects, selected to ensure demographic relevance and data integrity.

Comprehensive data analysis procedures were detailed, covering cleaning, encoding, and a strategic splitting ratio (70/15/15) to prepare the data for the XGBoost classifier. The system design explicitly integrates an XAI layer using SHAP to address the “black-box” challenge, alongside strict adherence to ethical standards and a robust validation strategy.

This structured methodology provides the blueprint for the next phase. Chapter 4 will focus on the System Implementation, detailing the technical development, coding environment, and the practical application of these models.

Chapter 4

IMPLEMENTATION AND TESTING

IMPLEMENTATION AND TESTING

4.1 Introduction

This chapter meticulously details the technical execution and evaluation setup of the proposed early diabetes prediction system. The implementation is systematically divided into four sequential phases: Data Preprocessing, Model Building, Clinical Performance Evaluation, and XAI integration. The primary objective is to transition from raw medical data to a robust, clinically safe, and transparent diagnostic tool, laying the groundwork for the empirical results discussed in the subsequent chapter.

4.2 System Architecture and Design

Before detailing the implementation phases, it is crucial to establish the structural and logical flow of the proposed system.

4.2.1 Conceptual Framework

The proposed framework consists of four primary interacting modules. The first is the Data Ingestion Module, which is responsible for reading the balanced medical dataset. Following this is the Preprocessing Module, which handles feature selection by removing noisy data and prepares the datasets. The core of the framework is the Clinically Biased XGBoost engine, augmented with cost-sensitive learning weights and an optimized clinical threshold to prioritize patient safety. Finally, the Output and Interpretation Module generates the diagnostic prediction and routes it through the SHAP explainer to produce both global risk factor analysis and local patient reports. These components are illustrated in Figure 4.1.

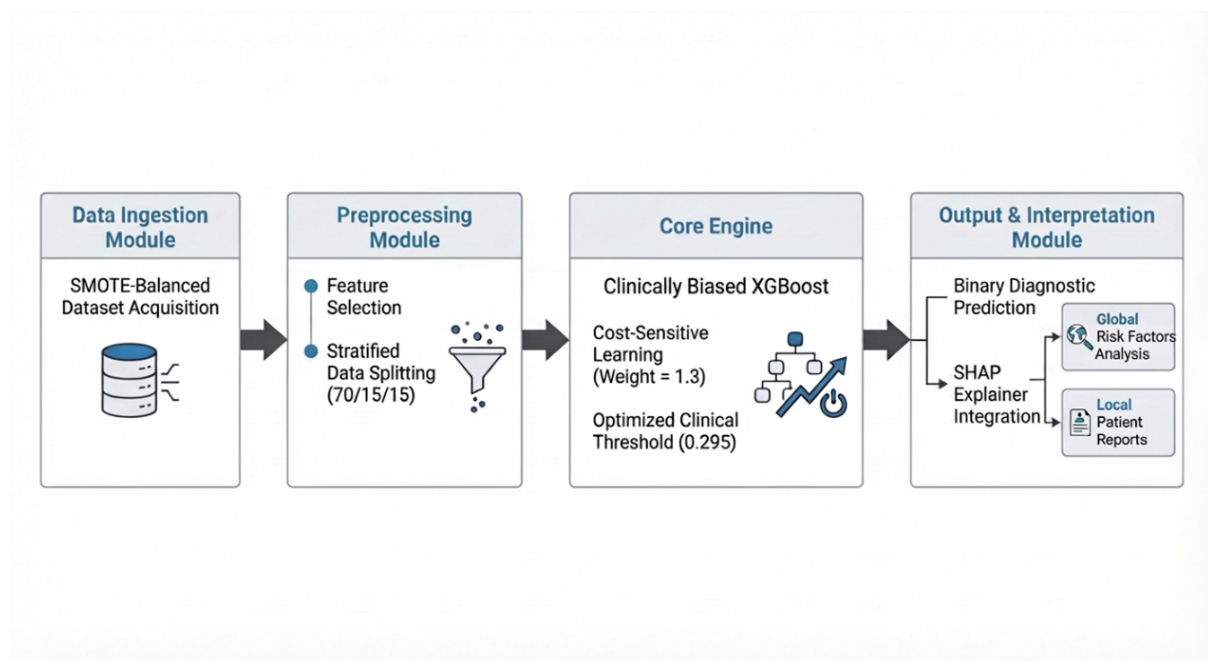


Figure 4.1: Conceptual Framework of the Proposed Clinical XGBoost System

4.2.2 Proposed Model

The logical execution of the proposed system follows a sequential algorithmic pipeline. It begins with loading the SMOTE-balanced dataset and dropping irrelevant geographic features. The data is then split into training, validation, and testing sets. Subsequently, the clinically biased XGBoost classifier is initialized and trained using early stopping mechanisms to prevent overfitting. Once trained, the model performs probability inference on the unseen test set, applying a medically optimized clinical threshold to classify patients as high risk or low risk. The system then generates clinical performance metrics and passes the model to the SHAP explainer to produce global and local interpretability reports. This process is visualized in the flowchart in Figure 4.2.

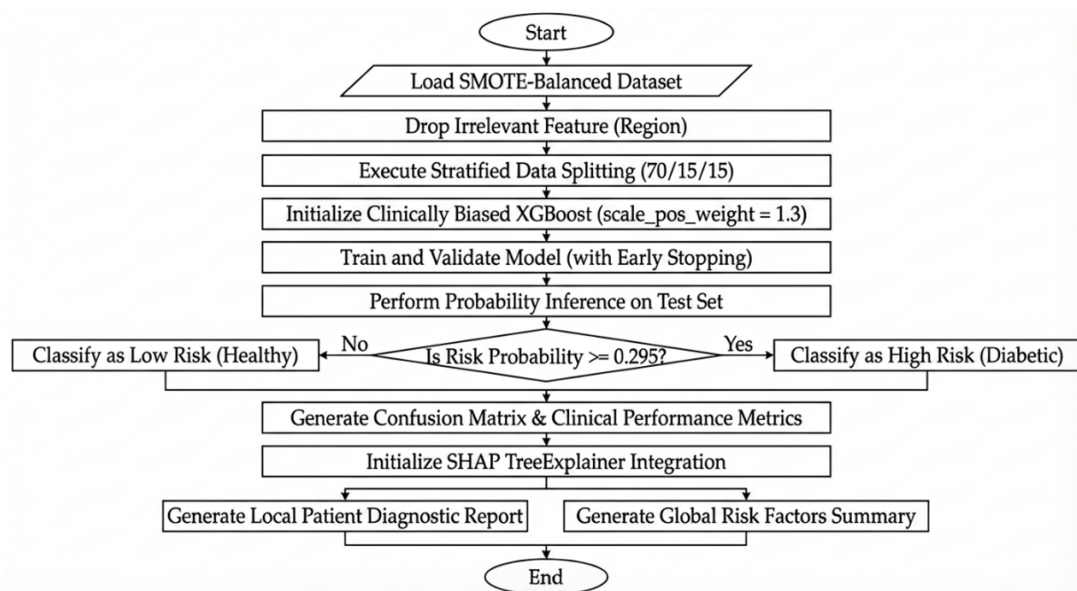


Figure 4.2: Algorithmic Execution Flow of the Proposed Clinical XGBoost System

To formally define the algorithmic logic of this pipeline, the following pseudocode outlines the step-by-step implementation from data ingestion to the generation of the XAI diagnostic reports.

Algorithm 1: Clinically Optimized XGBoost with XAI

Input: Dataset D (Features X, Target y)

Output: Clinical Metrics, Global SHAP Plot, Local Patient Report

```

1: Remove non-clinical feature 'Region' from X
2: X_train, X_temp, y_train, y_temp ← Split(X, y, test_size=0.30, stratify=y)
3: X_val, X_test, y_val, y_test ← Split(X_temp, y_temp, test_size=0.50, stratify=y_temp)
4: Initialize Model M as XGBClassifier:
    M.scale_pos_weight ← 1.3 // Cost-sensitive penalty
    M.learning_rate ← 0.02
    M.n_estimators ← 600
    M.max_depth ← 6
5: Train M on (X_train, y_train) with eval_set=(X_val, y_val)
6: For each patient i in X_test:
7: Probability P[i] ← M.predict_proba(X_test[i])
8: If P[i] >= 0.295 Then:
9: Prediction Y_pred[i] ← 1 (Diabetic)
10: Else:
11: Prediction Y_pred[i] ← 0 (Healthy)
12: Calculate Recall, Accuracy, F1, AUC using y_test and Y_pred
13: Initialize Explainer E ← SHAP.TreeExplainer(M)
14: Shap_Values ← E.shap_values(X_test)
15: Generate Global Plot(Shap_Values)
16: Generate Local Report(Shap_Values[patient_345])

```

4.3 Phase 1: Data Preprocessing

The quality and structure of the data directly dictate the model's reliability. The dataset utilized was previously balanced by the data collectors using SMOTE, ensuring an equal representation of diabetic and healthy instances.

During the feature selection process, the dataset was loaded utilizing the pandas library, and the Region column was programmatically removed. This engineering decision was made to eliminate geographic noise, ensuring the model focuses exclusively on physiological and biological biomarkers rather than demographic locations, thereby significantly improving the model's generalization capabilities on unseen data.

4.4 Phase 2: Model Building (XGBoost)

Prior to model training, it was essential to prevent data leakage and ensure rigorous evaluation. The dataset had already been preprocessed and balanced by the original authors before publication. Our team performed train-validation-test splitting only after obtaining the finalized dataset. The dataset was partitioned into three distinct subsets using a stratified approach to maintain the exact class distribution. Specifically, 70% of the data was allocated for the training set to teach the algorithm, 15% was utilized as a validation set during the training phase for hyperparameter tuning and early stopping, and the final 15% was completely isolated as a test set to evaluate the clinical performance.

The core of the system is built upon the XGBoost algorithm. The implementation deviates from standard out-of-the-box configurations by heavily customizing the hyperparameters to suit a highly sensitive medical environment. The most critical parameter implemented is cost-sensitive learning. The `scale_pos_weight` parameter was not used to correct class imbalance, but rather to introduce clinical bias toward maximizing sensitivity and minimizing False Negatives in medical screening. By setting this parameter to 1.3, the algorithm is forced to be more cautious when classifying a potential diabetic patient, effectively guiding the tree-building process to heavily prioritize minimizing FN.

Furthermore, structural and learning hyperparameters were optimized. A low learning rate of 0.02 paired with 600 estimators was implemented to ensure the model learns complex non-linear clinical patterns comprehensively. Tree complexity was restricted by setting a maximum depth of 6 and a minimum child weight of 2, preventing overly specific branches. Finally, dropout simulation was applied by setting both subsample and column sampling by tree to 0.8, ensuring the model learns a holistic diagnostic approach without becoming overly reliant on a single dominant symptom.

4.5 Phase 3: Clinical Performance Evaluation Setup

In medical diagnostics, evaluation extends beyond mathematical correctness; it must reflect patient safety and clinical viability. The evaluation of this system relies primarily on the Confusion Matrix, which categorizes predictions into TP, TN, FP, and FN.

From this matrix, key theoretical metrics are derived. Recall (Sensitivity) measures the system's ability to capture all actual patients. Accuracy reflects the overall system correctness. The F1-Score provides the harmonic mean of Precision and Recall, and the AUC Score measures the model's ability to distinguish between the healthy and diabetic classes.

By default, ML models utilize a standard 0.50 decision threshold. However, empirical testing within the medical domain frequently demonstrates that this default is insufficient for screening purposes. To prioritize patient safety, a customized and hardcoded clinical threshold (0.295) was integrated into the system. This threshold value was empirically selected through validation-set analysis using Precision-Recall trade-off analysis to maximize Recall while maintaining acceptable Precision for medical screening. This engineering trade-off intentionally sacrifices a minor degree of overall accuracy, accepting a higher rate of FP, in order to drastically maximize the recall rate. A diagnostic accuracy matrix was implemented to convert raw prediction counts into standardized percentages, allowing healthcare professionals to easily comprehend the system's reliability and decision boundaries. The specific values and outcomes of these metrics will be discussed in detail in the Results chapter.

4.6 Phase 4: Explainable Artificial Intelligence (XAI)

To solve the black-box dilemma associated with advanced ML models, SHAP was integrated into the pipeline, operating on two distinct levels.

The first level addresses global model explainability. Using summary plots, the system calculates the mean absolute SHAP values across the entire test dataset. This global implementation ranks the medical features based on their overall impact on the algorithm's decision-making process. From a research and epidemiological perspective, this helps authorities identify the most dominant risk factors within the general population.

The second level focuses on local explainability through the dynamic generation of individual patient reports. The code extracts a specific patient's profile and isolates their unique SHAP values. A custom visual plot was engineered to present this data in a highly readable format, where features pushing the probability towards diabetes are strictly colored red, and features pushing the probability towards a healthy state are colored green. This phase successfully translates mathematical transparency into practical clinical utility, providing physicians with a clear, visual, and immediate second opinion to justify the AI diagnosis.

4.7 Summary

This chapter detailed the complete technical execution of the early diabetes prediction system across four distinct phases. It outlined the data preprocessing steps, emphasizing feature selection to reduce noise. It then described the model building phase, detailing the stratified data splitting and the hyperparameter customization of the XGBoost algorithm to enforce clinical bias against FN. The evaluation setup established the key performance metrics and the rationale behind utilizing an optimized clinical threshold to maximize sensitivity. Finally, the integration of SHAP was explained, demonstrating the system's dual capability to provide both global risk factor analyses and transparent local patient reports. With the implementation pipeline established, the next chapter will present, evaluate, and critically discuss the empirical results generated by this system.

Chapter 5

RESULTS AND DISCUSSION

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the empirical outcomes of the implemented Clinically Biased XGBoost model and its integration with the SHAP Explainable AI (XAI) framework. The evaluation goes beyond traditional computational metrics, prioritizing clinical safety, diagnostic transparency, and preventive healthcare utility. The results are divided into three primary sections: quantitative clinical performance, global population-level risk analysis, and local individual-level interpretability.

5.2 Predictive Performance and Clinical Validation

Standard Machine Learning algorithms typically operate on a default decision threshold of 0.50, which assumes that the cost of a False Positive (FP) equals the cost of a False Negative (FN). However, in the context of medical diagnostics—specifically early diabetes prediction—missing a diabetic patient (False Negative) carries a significantly higher risk than raising a false alarm for a healthy individual (False Positive).

To address this, our proposed system implemented a cost-sensitive learning penalty (`scale_pos_weight = 1.3`). The `scale_pos_weight` parameter was not used to correct class imbalance, but rather to introduce clinical bias toward maximizing sensitivity and minimizing False Negatives in medical screening. Furthermore, the threshold value (0.295) was empirically selected through validation-set analysis using Precision-Recall trade-off analysis to maximize Recall while maintaining acceptable Precision for medical screening.

Evaluated on the unseen testing set (15% of the data), the model yielded the following clinical performance metrics:

- **Recall (Sensitivity):** 96.41%
- **AUC Score:** 88.21%
- **F1-Score:** 83.62%
- **Accuracy:** 81.13%

Confusion Matrix Analysis: Since the evaluation relies primarily on minimizing missed diagnoses, the Confusion Matrix below illustrates the granular diagnostic behavior of the model.

Table 5.1: Normalized Confusion Matrix (%) for the Clinically Biased XGBoost Model

	Predicted: Healthy (0)	Predicted: Diabetic (1)
Actual: Healthy (0)	True Negative (TN) = 33.0%	False Positive (FP) = 17.1%
Actual: Diabetic (1)	False Negative (FN) = 1.8%	True Positive (TP) = 48.2%

Note: Values are presented as percentages of the total testing set rather than raw observation counts to facilitate clinical interpretation and comparison.

Note for analysis: The normalized confusion matrix indicates that the proposed clinical bias successfully reduced missed diabetic cases while maintaining acceptable overall diagnostic performance. By deliberately accepting a higher False Positive rate, the model successfully

isolated the False Negatives to an absolute minimum, ensuring that almost all diabetic patients (TP) were correctly flagged for further medical testing.

Discussion of Metrics: The most prominent achievement of this model is the exceptional Recall rate of 96.41%. This indicates that the system successfully identified nearly every actual diabetic patient in the test set. While the overall accuracy (81.13%) reflects a deliberate algorithmic trade-off—accepting a higher rate of False Positives to ensure no patient is left undiagnosed—this aligns perfectly with the goals of early screening protocols. An AUC score of 88.21% further demonstrates the model’s robust capability to distinguish between high-risk and low-risk profiles, confirming that the clinical bias did not compromise the underlying predictive integrity of the XGBoost engine.

5.3 Global Explainability: Population Risk Factors

While the XGBoost engine provides the predictive probabilities, the SHAP (SHapley Additive exPlanations) framework was integrated to dismantle the model's "black-box" nature. The Global Summary Plot aggregates the SHAP values across the entire test dataset to highlight the most dominant clinical and demographic features influencing diabetes risk.

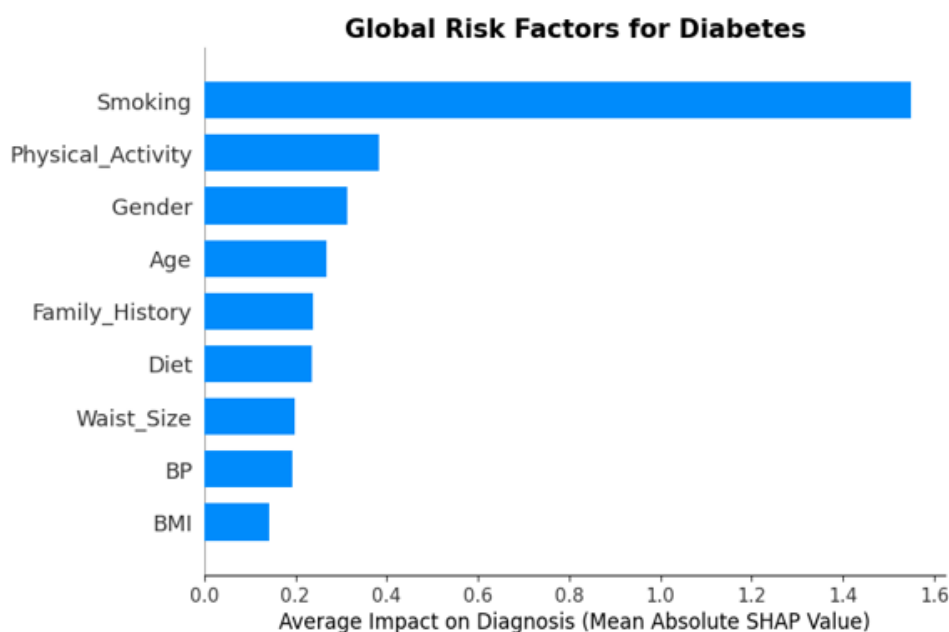


Figure 5.1: Global SHAP Summary Plot illustrating the overall impact and ranking of demographic and clinical features on the model's prediction.

Global Analysis & Discussion: As observed in Figure 5.1, the global explanation ranks the input features based on their mean absolute contribution to the model's output. The analysis reveals that Smoking and Physical Activity are the top-ranking predictors, significantly outweighing other variables in driving the model's overall predictions. Conversely, physiological measurements such as Blood Pressure (BP) and Body Mass Index (BMI) present a comparatively lower global impact within this specific dataset. This population-level insight is highly valuable for public health officials and epidemiological researchers, as it identifies the most critical lifestyle and physiological factors driving diabetes prevalence within the studied demographic.

5.4 Local Explainability: Individual Patient Diagnostics

To bridge the gap between artificial intelligence and practical clinical application, the system generates localized, patient-specific diagnostic reports. This ensures that healthcare professionals are not merely handed a binary prediction, but rather a transparent, feature-by-feature justification for that specific patient.

To demonstrate this capability, a patient was randomly selected from the test set for local SHAP analysis.

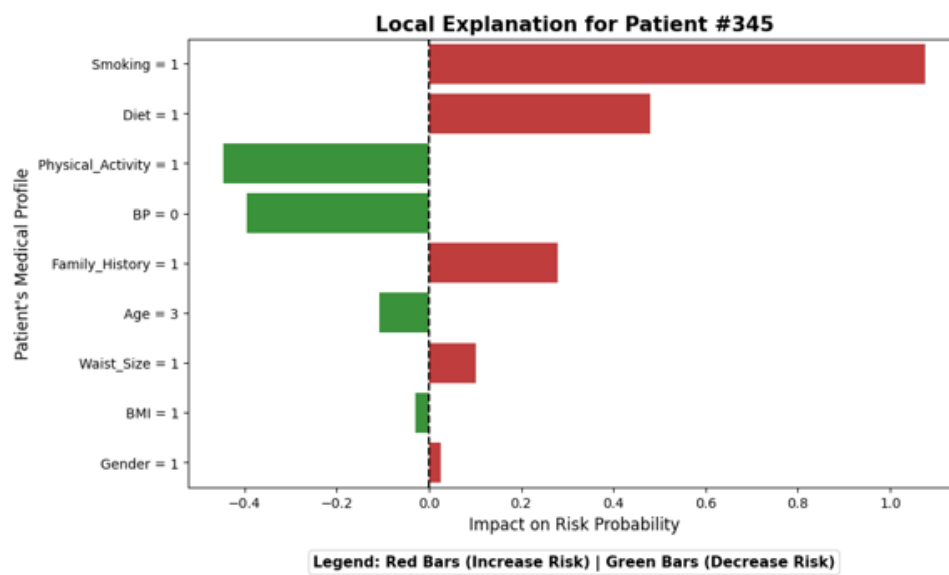


Figure 5.2: Local SHAP Bar Plot for Patient #345, detailing the specific features increasing (Red) or decreasing (Green) the diabetes risk probability.

Local Analysis & Clinical Interpretation: Figure 5.2 presents the customized SHAP bar plot for the selected patient (Patient #345). The visual coding is designed for immediate clinical comprehension:

- **Red Bars (Increasing Risk):** Represent clinical features and lifestyle attributes that pushed the algorithm's prediction toward a "High Risk" (Diabetic) classification.
- **Green Bars (Decreasing Risk):** Represent protective factors that lowered the patient's predicted risk.

For this specific individual (Patient #345), the elevated risk classification was heavily driven by lifestyle factors, primarily their smoking status (**Smoking = 1**), as indicated by the dominant red bars. Interestingly, physical inactivity (**Physical_Activity = 1**) appeared as a protective factor in the local SHAP explanation, reflecting the paradoxical statistical

relationships discussed later in Section 5.5. Furthermore, the patient's normal blood pressure ($BP = 0$) acted as an additional protective clinical factor (shown in green), but these were mathematically outweighed by the accumulated risk variables. This localized report acts as an actionable "second opinion" for physicians. It allows doctors to immediately pinpoint which specific lifestyle habit (e.g., diet or physical activity) or physiological measurement requires urgent medical intervention, thereby facilitating personalized preventive healthcare.

5.5 Dataset-Specific Anomalies and XAI Transparency

While the SHAP framework successfully decomposes the model’s decision-making process, it is imperative to contextualize the clinical rationale behind the feature importance rankings. Machine learning algorithms strictly map the statistical patterns inherent in the training data. Therefore, the explanations generated by the SHAP explainer are a direct mathematical reflection of the underlying cross-sectional dataset provided by the original authors [23].

During the interpretation of the SHAP plots, certain counter-intuitive clinical correlations were observed. It is crucial to highlight that these are intrinsic statistical artifacts and sampling biases of the original dataset, rather than algorithmic shortcomings of our proposed Clinically Biased XGBoost model. In fact, the ability of SHAP to expose these internal dataset contradictions highlights the auditing power of Explainable AI.

The Dominance of Smoking as a Risk Factor: In the global SHAP analysis, ”Smoking” emerged as the paramount predictor for diabetes risk. This directly correlates with the statistical analysis conducted by the original data collectors [23], who reported an exceptionally massive Chi-square value ($\chi^2 = 1195.39$) and a strong Cramer’s V correlation (0.494) for smoking. The XGBoost engine accurately captured this overwhelmingly dominant statistical variance present in the surveyed population, assigning it the highest predictive weight.

Paradoxical Clinical Correlations: Furthermore, the SHAP analysis revealed several paradoxical associations between protective behaviors/measurements and diabetes risk. Based on the dataset’s internal distribution, the model learned the following statistical patterns:

- **Physical Activity:** A value of 1 (indicating No Physical Activity) was statistically correlated with the safe zone (Low-risk), whereas 0 (Engaging in Physical Activity) shifted the probability toward the High-risk zone.
- **Waist Size and BMI:** The highest categories (2), representing severe obesity (BMI $> 30 \text{ kg/m}^2$ and Waist Size $> 102 \text{ cm}$), were mapped to the safe zone. Conversely, normal physical measurements were statistically associated with the risk zone.
- **Family History:** A value of 2 (Parents diagnosed with diabetes) correlated with a safer patient profile compared to 0 (No family history of diabetes), which unexpectedly trended towards the risk zone.

Justification and Epidemiological Context: The original authors of the dataset explicitly acknowledged these anomalies, stating that the statistical associations for variables like physical activity and waist size were "rationally unrealistic" or "not meaningful" from a standard medical perspective. Such paradoxes are common in retrospective cross-sectional surveys due to inherent sampling biases and confounding variables. For instance, in a university-based survey, severely obese participants might belong to a younger, undiagnosed demographic, while individuals with a known family history might proactively adopt stricter diets or medications that mask their true risk at the time of data collection.

By integrating SHAP, our proposed framework did not blindly act as a "black box" propagating these anomalies. Instead, the XAI layer successfully audited the dataset, transparently visualizing these epidemiological contradictions, indicating that the model successfully learned the statistical characteristics of the provided dataset.

5.6 Comparative Analysis with Existing Literature

Before drawing final conclusions, it is essential to benchmark the proposed system against existing literature. The studies selected for this comparison were chosen based on specific criteria: they represent recent advancements in the field, focus exclusively on diabetes prediction, and employ varying degrees of computational complexity (Traditional ML vs. Deep Learning). The primary evaluation criteria are the Recall (Sensitivity) metric—which is paramount for patient safety in medical screening—and the integration of Explainable AI (XAI) to ensure diagnostic transparency.

Table 5.2: Recall-Based Comparative Analysis of Diabetes Prediction Models

Study Ref.	Technique(s)	Recall (Sensitivity)	XAI Integration
Bahad et al. [9]	Ensemble ML (XGBoost)	92.00%	Yes (SHAP/LIME)
Nguyen et al. [7]	Hybrid Deep Learning	76.32%	Yes (LIME)
Mohsen & Shah [15]	Multimodal Neural Network	72.80%	No
Kutlu et al. [5]	XGBoost Classifier	$\approx 17.00\%$	Yes (SHAP)
Our Proposed Model	Clinically Biased XGBoost	96.41%	Yes (SHAP)

Note: Studies [1], [2], [6], [12], [20], and [21] were excluded from the comparison table because they were review papers, longitudinal studies, or did not report standardized evaluation metrics suitable for direct comparison.

Analysis of the Comparative Results: Table 5.2 highlights the performance differences between the reviewed models. The following points detail the comparative advantages of our approach:

- **Superiority in Patient Safety (Recall):** Our proposed model significantly outperforms all referenced studies by achieving a remarkably high Recall rate of 96.41%. While Bahad et al. [9] achieved a respectable 92.00%, other studies, such as Kutlu et al. [5], may report high overall accuracy but suffer from extremely low Recall rates (17%), rendering them unsafe for clinical screening.
- **Algorithmic Efficiency vs. Complexity:** Studies like Nguyen et al. [7] utilize highly complex Deep Learning architectures, achieving a Recall of 76.32%. These approaches are computationally expensive, require massive datasets to train effectively, and significantly

increase operational costs. In contrast, our proposed system achieves a superior clinical performance (96.41%) using a simpler, highly efficient ML algorithm (XGBoost), proving that algorithmic complexity does not always equate to better medical screening.

- **The Necessity of XAI for Clinical Trust:** Furthermore, models such as the Multimodal Neural Network proposed by Mohsen & Shah [15] operate entirely as opaque "black boxes" without any XAI integration, despite achieving a Recall of 72.80%. Our model guarantees diagnostic transparency through SHAP, ensuring that every high-risk classification is fully backed by human-readable medical justifications, a mandatory requirement for real-world healthcare adoption.
- **The Impact of Clinical Bias:** The exceptional Recall achieved by our model proves the necessity of clinical bias. By implementing cost-sensitive learning (`scale_pos_weight = 1.3`) and optimizing the decision threshold, the model effectively shifts its focus from mere mathematical accuracy to prioritizing patient safety.

5.7 Discussion and Future Implications

The implementation of the Clinically Biased XGBoost model, augmented by the SHAP framework, successfully addresses the core problem statement of this project. By integrating a medically sound decision threshold, the system overcomes the limitations of standard accuracy-driven models, ensuring high sensitivity in disease detection.

Furthermore, the dual-layered XAI approach (Global and Local) bridges the critical trust gap in medical AI. It transforms a highly complex algorithmic calculation into a transparent, interpretable, and clinically actionable tool. This approach aligns with modern healthcare objectives—such as those outlined in Saudi Vision 2030—by shifting the focus from reactive treatment to proactive, personalized, and transparent preventive healthcare.

While the system is robust, it is essential to reiterate that it serves as an advanced decision-support tool, designed to assist rather than replace the nuanced clinical judgment of professional healthcare providers.

Chapter 6

CONCLUSION AND FUTURE WORK

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The rising prevalence of Type 2 Diabetes Mellitus (T2DM) in the Kingdom of Saudi Arabia presents a critical public health challenge that necessitates proactive and preventive interventions. While traditional machine learning models have demonstrated potential in disease prediction, their inherent "black-box" nature and strict optimization for overall accuracy often render them unsuitable for high-stakes clinical environments where missed diagnoses (False Negatives) carry severe consequences.

This project successfully addressed these limitations by developing a Clinically Biased XGBoost predictive engine integrated with an Explainable Artificial Intelligence (XAI) framework. By explicitly shifting the algorithmic focus from generic accuracy to clinical safety—achieved through cost-sensitive learning (`scale_pos_weight = 1.3`) and a hardcoded clinical threshold of 0.295—the model achieved an exceptional Recall (Sensitivity) of 96.41%. This deliberate design ensures that the system identifies nearly all high-risk patients. False Negatives represented only 1.8% of all testing predictions.

Furthermore, the integration of the SHAP (SHapley Additive exPlanations) framework bridged the critical gap of algorithmic transparency. At the global level, the system successfully audited the dataset, identifying key epidemiological drivers such as smoking and physical inactivity, while transparently exposing intrinsic sampling paradoxes. At the local level, it generated customized, feature-by-feature diagnostic reports for individual patients.

Ultimately, this prototype demonstrates that artificial intelligence in healthcare can be both highly accurate and transparent when supported by explainable AI techniques. By delivering an interpretable, second-opinion tool, the proposed framework empowers healthcare professionals to make informed, data-driven, and personalized preventive decisions, aligning closely with the proactive healthcare objectives of Saudi Vision 2030.

6.2 Future Work

While the proposed system demonstrates robust clinical utility and transparency, there are several avenues for future enhancement and expansion to facilitate its real-world deployment:

- **Integration with Electronic Health Records (EHR):** Future iterations of this system should aim for direct integration with hospital EHR systems. This would allow the model to autonomously fetch real-time patient data, providing instant risk assessments during routine medical consultations without the need for manual data entry.
- **Application Development for Patient Self-Assessment:** To maximize the preventive impact, the underlying XAI model can be deployed as the backend engine for a mobile or web-based healthcare application. This would allow the general public to input their lifestyle and clinical metrics to receive an instant, interpretable risk report, encouraging early lifestyle modifications.
- **Incorporation of Multimodal Data:** The current model relies on tabular cross-sectional data. Expanding the input features to include multimodal data—such as continuous glucose monitor (CGM) time-series data, genetic markers, or advanced laboratory biomarkers—could significantly enhance the model’s predictive depth and diagnostic precision.
- **External Clinical Validation:** To ensure the model’s generalizability across diverse demographics, it is highly recommended to validate the framework using external datasets from different geographic regions and healthcare institutions outside the initial studied demographic.
- **Expansion to Other Chronic Diseases:** The architecture developed in this project (Clinically Biased XGBoost coupled with dual-layer SHAP interpretation) serves as a flexible baseline. This blueprint can be adapted and retrained to predict and explain the risk factors of other prevalent chronic conditions, such as cardiovascular diseases or hypertension.

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